

PROJECT PLANNING DOCUMENT

India's Agricultural Crop Production Analysis using Tableau

Internship Platform:	SmartBridge
Tool Utilized:	Tableau Public
Dataset:	Indian Agricultural Crop Production Dataset (1997–2021)
Submission Date:	13 July 2026

1. AGILE ESTIMATION & PROJECT BACKLOG

To ensure systemic tracking and execution, the lifecycle activities are mapped to defined Epics and User Stories using an effort estimation model. The calculation yields a balanced workflow capacity across the distribution timeline.

- **Total Estimated Story Points:** 32 Points
- **Total Scheduled Duration:** 2 Sprints
- **Calculated Team Velocity:** 16 Story Points per Sprint

SPRINT	EPIC	USER STORY DESCRIPTION	STORY POINTS
Sprint 1	Data Collection	USN1: Gathering Indian Agricultural Dataset (1997–2021)	2
Sprint 1	Data Collection	USN2: Loading Data into Tableau Public environment	1
Sprint 1	Data Preparation	USN3: Handling Missing Values & record validation	3
Sprint 1	Data Preparation	USN4: Creating "Yield" Calculated Field via custom formulas	3
Sprint 1	Data Preparation	USN5: Remediation of text inconsistencies in state names	3
Sprint 2	Data Visualization	USN6: Developing Crop Comparison Bar Charts	2
Sprint 2	Data Visualization	USN7: Building Distribution Pie Charts	2

SPRINT	EPIC	USER STORY DESCRIPTION	STORY POINTS
Sprint 2	Data Visualization	USN8: Developing Year-wise Production Line Charts	2
Sprint 2	Data Visualization	USN9: Creating Geographic State-wise Maps	4
Sprint 2	Dashboard	USN10: Developing Interactive Dashboard with KPI Cards	5
Sprint 2	Story	USN11: Developing Tableau Story for presentation narrative	5

2. PROJECT LIFECYCLE PHASES

Phase 1 – Problem Identification

- **Understand agricultural data analysis challenges:** Interrogating huge, historical crop data records makes identifying regional production imbalances difficult without visual aggregation tools.
- **Define objectives:** Build a clean, scalable platform highlighting crop distribution trends, cultivated surface areas, and calculated crop yields across diverse geographies.

Phase 2 – Requirement Analysis

- **Study dataset:** Audited primary column headers including state metadata, seasonal variables, baseline area sizes, and overall raw production volumes.
- **Identify required visualizations:** Pinned down geospatial choropleths for regional overviews, alongside detailed multi-axis line and bar distributions.
- **Plan dashboard structure:** Designed standard UX hierarchies establishing top summary indicators, core geographical plots, and comprehensive filtering sidebars.

Phase 3 – Data Collection

- **Obtain agricultural dataset:** Extracted longitudinal historical database tracking Indian crop yields over the 1997–2021 temporal spectrum.
- **Verify dataset integrity:** Executed validation scans to ensure format standardization and row continuity across variables.

Phase 4 – Data Cleaning & Preprocessing

- **Remove missing values:** Scrubbed empty records within the production metrics to prevent skewing systemic trends.

- **Validate records:** Dropped irrational row exceptions where reported dimensions violated baseline agricultural thresholds.
- **Create Yield calculated field:** Programmed calculated variables directly inside the analytics framework to isolate production output values relative to area sizes.

Phase 5 – Dashboard Development

- **Create worksheets:** Drafted focused visualization objects isolating standalone variables such as year-over-year adjustments.
- **Build state-wise maps:** Modeled dense spatial heatmaps representing state performance tiers using color gradients.
- **Create crop comparison charts:** Configured grouped dynamic bar matrices contrasting high-volume staples against seasonal cash products.
- **Add KPI cards:** Programmed distinct metric readouts detailing cumulative macro output and country-wide historical efficiency values.
- **Add filters:** Tied unified filter widgets across multiple worksheets to enable cross-filtering based on specific years, crop types, or distinct seasons.

Phase 6 – Story Development

- **Arrange dashboard sequence:** Mapped an analytical narrative progression descending from broad national summaries down to specific district insights.
- **Create Tableau Story:** Combined visualizations into an interactive presentation structure suitable for external reporting.

Phase 7 – Testing

- **Validate calculations:** Cross-checked computed yield results against manual calculation sets to guarantee system accuracy.
- **Test filters:** Conducted interface validation passes ensuring global filters propagate instantly across all active dashboard panels.
- **Check dashboard performance:** Inspected processing refresh lags to maintain snappy execution speeds during interaction loops.

Phase 8 – Publishing

- **Publish Dashboard:** Deployed target layout architectures directly onto the web platform.
- **Publish Story:** Uploaded presentation flows to secure cloud profiles.
- **Verify Tableau Public links:** Tested production URLs through neutral browser instances to confirm universal stakeholder access.

Phase 9 – Documentation

- **Prepare project documentation:** Consolidated configuration steps, operational structures, and technical workflow paths into formal guides.
- **Capture dashboard screenshots:** Documented states of dashboard objects under specific filtered conditions.
- **Final report preparation:** Packaged text metrics and graphical reviews into a final deliverable layout ready for submission.

3. PROJECT TIMELINE SCHEDULE

Phase / Task Track	W1	W2	W3	W4	W5	W6	W7	W8	W9	Status
Phase 1: Problem Definition	[XX]									Completed
Phase 2: Requirement Analysis		[XX]								Completed
Phase 3: Dataset Collection			[XX]							Completed
Phase 4: Data Cleaning				[XX]						Completed
Phase 5: Dashboard Development					[XX]					Completed
Phase 6: Story Development						[XX]				Completed
Phase 7: Testing							[XX]			Completed
Phase 8: Publishing								[XX]		Completed
Phase 9: Documentation									[XX]	Completed

4. PROJECT DELIVERABLES

DELIVERABLE DESCRIPTION	CURRENT LIFECYCLE STATUS
Problem Statement	Completed
Empathy Map	Completed
Brainstorming	Completed
Customer Journey Map	Completed
Solution Requirement	Completed
Technology Stack	Completed
Dashboard	Completed
Story	Completed
Tableau Public Link	Completed

DELIVERABLE DESCRIPTION	CURRENT LIFECYCLE STATUS
Documentation	Completed
Final Report	Completed

5. MILESTONES ACHIEVED

MILESTONE TARGET	DESCRIPTION	DATE COMPLETED
Problem Defined	Analytical operational issues, scope, and parameters fully established.	13 July 2026
Dataset Ready	Historical matrices acquired, standardizations applied, and metrics computed.	13 July 2026
Dashboard Completed	Worksheets, interactive maps, comparison metrics, and filtering tied down.	13 July 2026
Story Completed	Logical analytical narrative sequences compiled inside presentation elements.	13 July 2026
Published to Tableau Public	Project files deployed live and remote hyperlinked endpoints fully verified.	13 July 2026
Documentation Completed	System architectures, code properties, and interface states thoroughly logged.	13 July 2026
Project Submitted	All technical analytics components structured and passed for final review.	13 July 2026

6. RISK ASSESSMENT & MITIGATION MATRIX

IDENTIFIED RISK FACT	APPLIED SYSTEMIC MITIGATION STRATEGY
Missing Data	Deployed filter conditions to filter incomplete records without affecting historical trends.
Incorrect Calculations	Validated custom calculation models against independent calculation baselines.
Visualization Errors	Utilized industry-standard templates and distinct color scales to maximize visibility.

IDENTIFIED RISK FACT	APPLIED SYSTEMIC MITIGATION STRATEGY
Publishing Issues	Maintained localized workbook instances to avoid server upload drops.
Internet Connectivity	Configured local offline execution datasets to ensure project flow continuity.

7. CONCLUSION

Proper project planning ensured systematic development, timely completion, and successful implementation of the Tableau-based agricultural analysis project. Organizing tasks sequentially across distinct engineering blocks—from targeted data processing up through final dashboard assembly—minimized structural delivery friction and ensured complete baseline coverage. The completed web tool handles extensive datasets seamlessly and delivers reliable analytics. The project package stands optimized and ready for formal SmartBridge submission on **13 July 2026**.